

TeamPulse — MERN Full-Stack Learning Manual

Purpose

Ye PDF project ko sirf chalane ke liye nahi, samajhne ke liye hai. Target ye hai ki course/project complete hone ke baad tum bata sako: kya use hua, kyun use hua, frontend se backend tak request kaise gayi, MongoDB mein data kaise store hua, AI kaise connect hua aur privacy kaise enforce hui.

1. MERN ka complete flow

React UI → API request → Express route → JWT middleware → business logic/service → Mongoose model → MongoDB → JSON response → React state → UI.

2. Technology ka kaam

React: UI. Vite: frontend development/build. Node.js: server runtime. Express: REST API framework. MongoDB: database. Mongoose: MongoDB models/queries. JWT: authentication. bcrypt: password hashing. Claude: server-side AI generation.

3. Important features

Auth + teams, daily steps/sleep tracking, manual entry, seeded demo data, AI team challenge, Habit Rescue, challenge feedback, adaptive target loop, privacy-safe HR dashboard, 5-member anonymity rule, small noise on aggregates, HR audit log, impact report and ROI estimator.

4. Frontend ko kaise samjho

App.jsx screen composition aur state handle karta hai; api.js authenticated HTTP requests ko centralize karta hai; styles.css visual system/responsive layout rakhta hai. Screen ko component → state → API call → response → UI update ke flow mein padho.

5. Backend ko kaise samjho

app.js routes register karta hai. middleware.js JWT aur HR authorization check karta hai. models.js MongoDB schemas define karta hai. services.js reusable business logic rakhta hai. routes folder actual API endpoints rakhta hai.

6. Example: steps save karna

User steps enter karta hai → React POST /api/metrics bhejta hai → JWT verify hota hai → values validate hoti hain → Mongoose upsert se user+date record create/update hota hai → MongoDB save karta hai → JSON response → React week data reload karta hai.

7. Authentication

Signup par password bcrypt se hash hota hai. Login par bcrypt compare hota hai. Successful login par JWT issue hota hai. Protected requests Authorization Bearer token bhejti hain. Middleware token verify karke user identity route ko deta hai.

8. AI architecture

Browser mein Claude API key nahi hoti. React apne Express endpoint ko call karta hai. Server team-level averages/context Claude ko deta hai. JSON validate hota hai. Claude fail ho to safe fallback challenge/rescue message use hota hai.

9. Privacy architecture

HR endpoint raw user metrics return nahi karta. Team averages calculate hote hain. 5 se kam members wali team hide hoti hai. Eligible aggregates mein small random noise add hota hai. HR overview access audit log mein record hota hai.

10. Health Connect

Web MERN app Android Health Connect ko directly access karne ka claim nahi karta. Manual entry aur seeded demo data available hain. Real Health Connect ke liye native Android bridge/client chahiye. Is boundary ko interview mein honestly explain karna hai.

11. AI learning loop

Challenge result mein completion percentage aur thumbs feedback store hota hai. Latest three results dekhe jaate hain. Average completion >80% ho to next target ~10% badh sakta hai; <40% ho to ~10% kam ho sakta hai. Ye explainable feedback rule hai, fake machine-learning claim nahi.

12. HR ROI

Annual program cost = employees × monthly price × 12. Break-even estimate = annual program cost / estimated resignation cost, rounded up. Isse estimate ke roop mein label kiya gaya hai, promise nahi.

13. Course ke end tak tumhe kya explain karna chahiye

MERN, HTTP, GET/POST, Express routes, middleware, JWT, bcrypt, MongoDB documents, Mongoose models, CRUD/upsert, React state/props, frontend-backend connection, server-side secrets, MongoDB aggregation, privacy boundaries, AI fallback, testing aur deployment.

14. Interview mein 30-second explanation

“TeamPulse ek MERN workplace wellbeing product hai. React employee aur HR interfaces handle karta hai, Express protected REST APIs deta hai, MongoDB users/teams/metrics store karta hai, aur Claude Node backend se call hota hai. Product ka important part privacy hai: HR ko sirf team-level aggregates milte hain, 5 se kam members wali teams hidden hain aur individual metrics HR UI ko nahi milte. Habit Rescue aur adaptive challenges product ko static dashboard se different banate hain.”

15. Hackathon quality checklist

Har route working; mobile + desktop responsive; loading/empty/error states; secure environment variables; no individual HR data; AI fallback; demo seed/reset; polished UI; real pilot numbers only; README + learning manual; project ko bina AI ke verbally explain kar paana.